Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

Technical Requirement ID: TR-SQG-001

Related Functional Requirement(s): FR-SQG-001: Generate SQL Query for Sales Orders

Objective: Implement a system to generate a complete SQL SELECT statement that applies transformation logic, includes appropriate joins, and ensures correct data types and formatting based on the provided tabular mapping document.

Target Cluster Configuration:

Cluster Name: SQL Query Generator Cluster

Databricks Runtime Version: 7.3 LTS

Node Type: Standard_DS3_v2

Driver Node: 1 x Standard_DS3_v2

Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)

Autoscaling: Enabled

Auto Termination: 30 minutes

Libraries Installed: Spark SQL, Scala

Source Data Details:

Dataset Name: tabular mapping document

Location: /mnt/mapping-document/tabular-mapping.csv

Format: CSV

Description: contains the required columns (Column Name, Column Type, Source Table, Source Column/Transformation Logic, Join Condition)

Dataset Name: source tables (e.g., b_source.ord_hdr, b_source.ord_dtl)

Location: various databases and tables

Format: various (e.g., Parquet, CSV)

Description: input data for generating SQL query

Target Data Details:

Output Name: generated SQL query

Location: /mnt/generated-sql-queries/sales-orders.sql

Format: SQL

Description: complete SQL SELECT statement for sales orders

Job Flow / Pipeline Stages:

Stage 1: Retrieve tabular mapping document for "sales_orders_comp_stg" and "sales_orders_comp"

Stage 2: Generate SQL SELECT statement for "sales_orders_comp_stg" and "sales_orders_comp" based on mapping document

Stage 3: Apply transformation logic and joins as specified in the mapping document

Stage 4: Format the generated SQL query for readability

Data Transformations / Business Logic:

Step: Create Intermediate Table "sales_orders_comp_stg"

Description: generate SQL SELECT statement for "sales_orders_comp_stg"

Transformation Logic: concatenate "referenceto" and "refno" to generate "CompOrderNumber", perform left join between "b_source.ord_hdr" and "b_source.ord_dtl" on "referenceto" and "refno"

Step: Create Main Table "sales_orders_comp"

Description: generate SQL SELECT statement for "sales_orders_comp"

Transformation Logic: retrieve data from "s_master.sale_orders" and apply transformation logic as specified, perform joins with other tables as specified

Error Handling and Logging:

- Log errors and exceptions during SQL query generation
- Store logs in /mnt/logs/sql-query-generator.log
- Notify administrators in case of critical errors